

Winning Space Race with Data Science

Name: Khayelihle Mfeka
Date: 17 July 2026



Outline

- Executive summary
- Introduction
- Methodology
- Results
- Conclusion

Executive summary

Summary of methodologies

1. Data Collection:

- SpaceX launch data was accessed through the API and supplemented with records obtained via web scraping from Wikipedia.

2. Data Cleaning & Preparation:

- Cleaned and formatted the data.
- Stored data in a database and performed SQL queries.
- Conducted exploratory data analysis.

3. Feature Engineering:

- Created new features and standardized the data.

4. Interactive Visualizations:

- Mapped launch sites and success rates using Folium.
- Built an interactive dashboard with Plotly Dash.

5. Model Building & Evaluation:

- Implemented SVM, Decision Trees, and K-Nearest Neighbors.
- Tuned hyperparameters with GridSearchCV.
- Evaluated models using test data accuracy.

Summary of all results

1. Data Insights:

- Identified factors influencing Falcon 9 first stage landings.
- Visualized geographical patterns and success rates.

2. Model Performance:

- Logistics Regression, SVM and K-Nearest Neighbors: 83.33% accuracy.
- Decision Tree: 88.89% accuracy.

3. Key Findings:

- Launch site and payload mass impact landing success.
- Decision Tree model is the most effective predictor.

Introduction

Project background and context:

This capstone project is dedicated to developing predictive models that assess the probability of a successful landing of the Falcon 9 first stage. SpaceX has distinguished itself within the aerospace industry by offering launch services at markedly reduced costs compared to traditional providers, a competitive advantage largely attributable to its pioneering capability of reusing the first stage of its rockets. The accurate prediction of landing outcomes is therefore not only a technical challenge but also a critical factor in estimating launch expenditures. By establishing reliable forecasts, this project aims to generate insights that can inform cost analyses and strategic decision-making for organizations competing with SpaceX in the commercial launch market.

Problems you want to solve:

- 1.Which variables and conditions determine the probability of a successful landing of the Falcon 9 first stage?
- 2.How can we accurately predict the landing outcome using machine learning models?
- 3.Which machine learning model performs best in predicting the landing success?

Section 1

Methodology

Methodology

Executive summary: This project adopts an integrated methodology to forecast the successful landing of the Falcon 9 first stage, encompassing data acquisition, preprocessing, exploratory analysis, interactive visualization, and predictive modeling.

Data collection methodology:

- The dataset was retrieved from the SpaceX API, which provides comprehensive records of Falcon 9 launches, including dates, sites, payloads, and outcomes.

Perform data wrangling:

- The data cleaning process resolved missing values, standardized formats, and ensured overall consistency. Key features were extracted, while additional features were engineered to enrich the dataset.

Perform exploratory data analysis (EDA) using visualization and SQL:

- Launch success rates, payloads, and launch sites were visualized using Matplotlib and Seaborn.
- SQL queries were executed to extract insights and address specific questions related to the dataset.

Perform interactive visual analytics using Folium and Plotly Dash:

- Folium was utilized to develop interactive maps that visualize launch sites and their corresponding outcomes.
- A Plotly Dash application was developed with interactive components, including dropdowns and sliders, to facilitate the analysis of launch success rates and payload ranges.

Methodology

Perform predictive analysis using classification models:

- Built and evaluated various classification models including SVM, KNN, and Decision Trees.
- Employed GridSearchCV for hyperparameter tuning.
- Evaluated models based on accuracy, and identified the best performing model for predicting landing success.

Link to GITHUB : [KhayelihleMfeka/khayalBM](https://github.com/KhayelihleMfeka/khayalBM)

Data collection

Step 1: Space API Request



Step 2: Web Scraping Wikipedia



Step 3: Data Integration



Data collection – SpaceX API

Step 1: Initiate API Request

- Use Python's `requests` library to connect to the SpaceX API.
- Endpoint: `https://api.spacexdata.com/v4/launches`

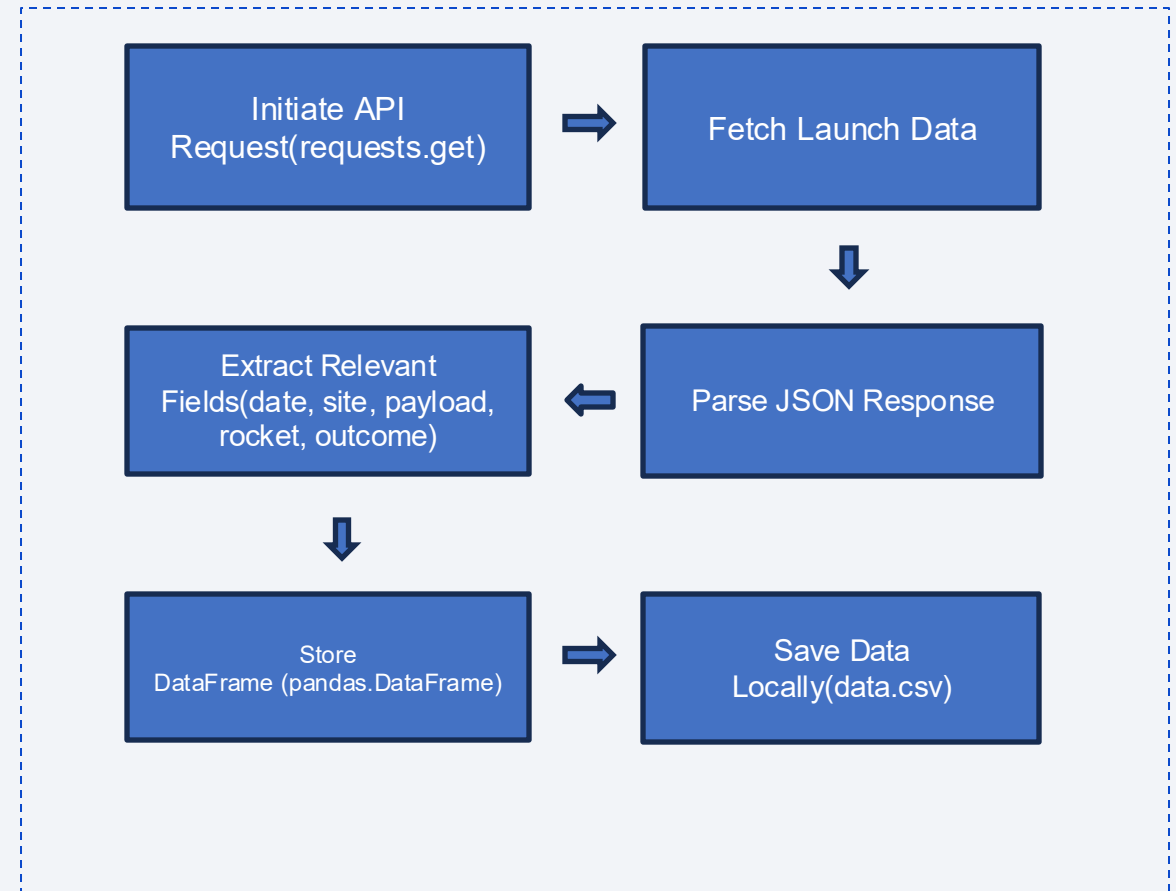
Step 2: Parse API Response

- Convert API response from JSON to a Python dictionary.
- Extract relevant fields: launch date, launch site, payload mass, rocket type, outcome.

Step 3: Store Data on local drive

- Save extracted data into a pandas DataFrame.
- Store the DataFrame locally for further processing.

GitHub URL: [khayalBM/1.jupyter-labs-spacex-data-collection.ipynb](https://github.com/khayalBM/1.jupyter-labs-spacex-data-collection.ipynb) at main · KhayelihleMfeka/khayalBM



Data collection – Scraping

Step 1: Initiate Web Scraping

- Use Python's `requests` library to fetch the HTML content of the Wikipedia page.
- Target URL: `https://en.wikipedia.org/wiki/List\_of\_Falcon\_9\_and\_Falcon\_Heavy\_launches`

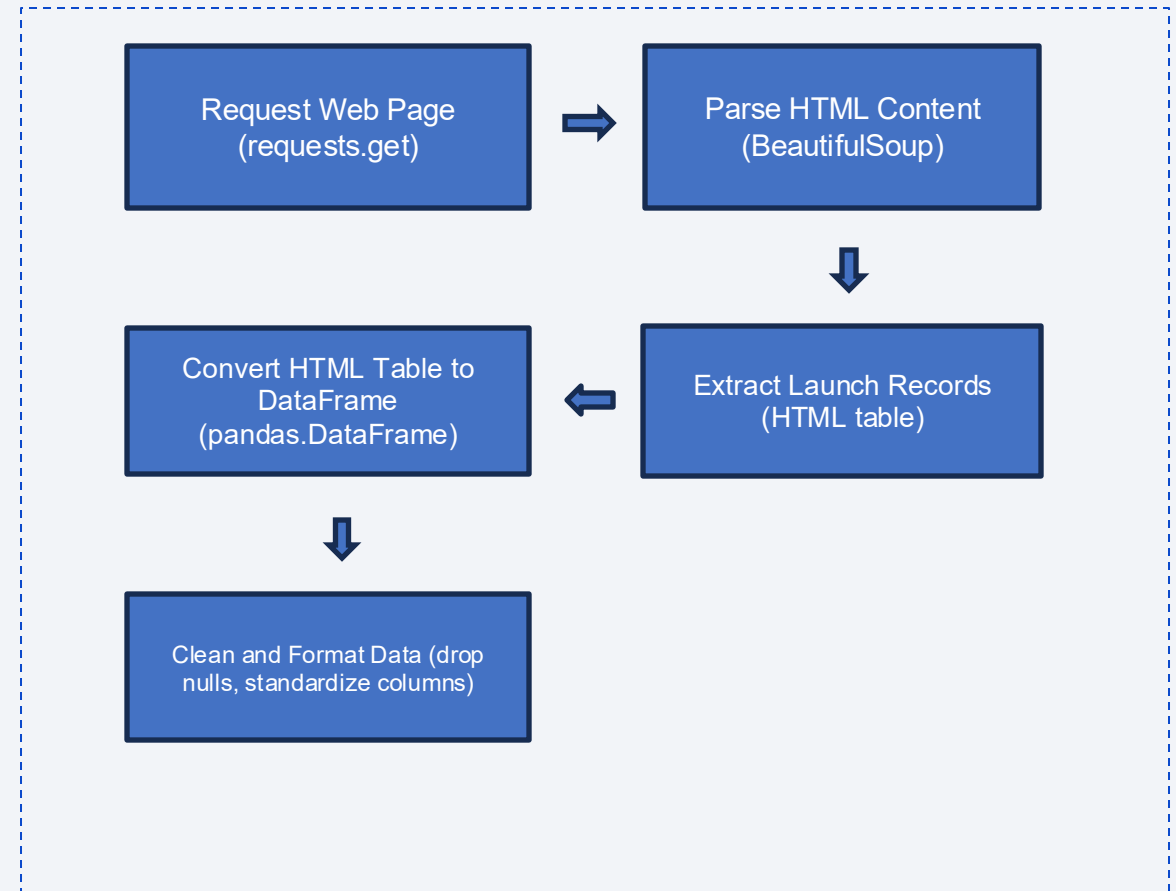
Step 2: Parse HTML Content

- Use `BeautifulSoup` to parse the HTML content.
- Extract the HTML table containing Falcon 9 launch records.

Step 3: Convert to DataFrame

- Convert the extracted HTML table into a pandas DataFrame.
- Clean and format the DataFrame, ensuring data consistency.

GitHub URL: [khayaIBM/2.jupyter-labs-webscraping.ipynb](https://github.com/KhayelihleMfeka/khayalBM/2.jupyter-labs-webscraping.ipynb) at main · KhayelihleMfeka/khayalBM



Data wrangling

Data wrangling entails cleaning, transforming, and structuring raw data into an organized format suitable for analysis.

Step 1: Data Cleaning

- Identify and fill or remove missing values in the dataset.
- Use appropriate imputation techniques or drop rows/columns with excessive missing data.

Step 2: Data Transformation

- Convert data types to appropriate formats (e.g., date-time, numerical).
- Standardize text (e.g., lowercase, remove whitespace).
- Create new features from existing data (e.g., extract year from date).
- Normalize/scale numerical features to ensure consistency.

Step 3: Data Integration

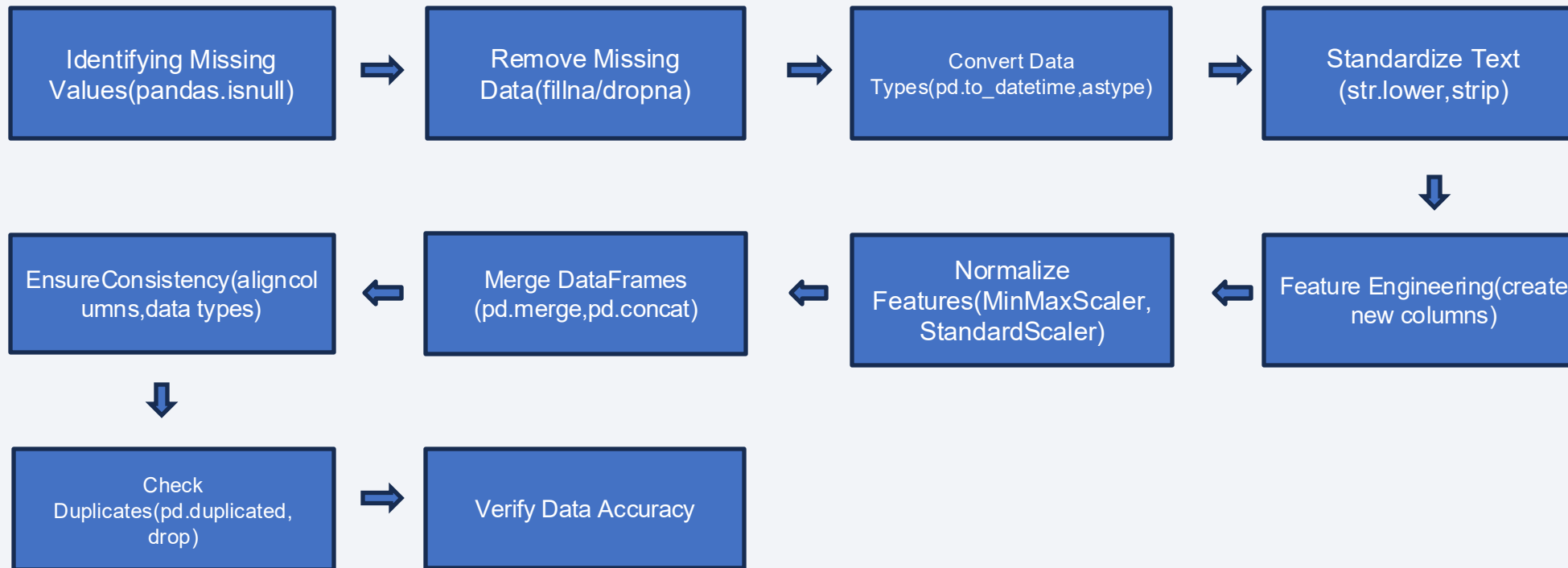
- Merge datasets collected from different sources (API, web scraping) into a single cohesive dataset.
- Ensure consistent column names and data formats across datasets.

Step 4: Data Validation

- Check for duplicate records and remove them.
- Verify the accuracy and consistency of data entries.

GitHub URL: [khayaIBM/3.labs-jupyter-spacex-data-wrangling_jupyterlite.ipynb](https://github.com/KhayelihleMfeka/khayaIBM/3.labs-jupyter-spacex-data-wrangling_jupyterlite.ipynb) at main · KhayelihleMfeka/khayaIBM

Data wrangling - Flowchart



EDA with data visualization

Exploratory Data Analysis (EDA) entails visually examining and summarizing the key characteristics of a dataset, with the objective of understanding its distribution, identifying patterns, and uncovering relationships among variables.

Charts Plotted:

1. Histograms:

- Purpose: Used to visualize the distribution of numerical variables such as launch success rates, payload mass, and flight number.
- Why: Helps in understanding the spread and central tendency of the data, identifying outliers, and assessing data skewness.

2. Bar Charts:

- Purpose: Used to compare categorical variables such as launch outcomes (success/failure) across different categories like launch sites or rocket types.
- Why: Provides a clear comparison of frequencies or proportions within categorical data, highlighting patterns or trends.

3. Line Charts:

- Purpose: Used to track trends over time, such as the success rate of Falcon 9 launches across different years.
- Why: Reveals temporal patterns and helps in understanding performance trends or changes over specific periods

4. Scatter Plots:

- Purpose: Used to explore relationships between two numerical variables, such as payload mass vs. launch success
- Why: Identifies correlations or dependencies between variables, visualizing how one variable changes concerning another.

5. Heatmaps:

- Purpose: Used to visualize correlation matrices between multiple numerical variables.
- Why: Helps in identifying strong correlations (positive or negative) between variables, aiding feature selection or understanding multicollinearity.

6. Box Plots:

- Purpose: Used to display the distribution of numerical data through their quartiles.
- Why: Visualizes the spread and skewness of data, highlighting outliers and comparing distributions across different categories.

EDA with SQL

Aggregate Queries:

- Calculated total number of launches.
- Counted successful and failed launches.
- Calculated success rates by launch site and rocket type.

Join Queries:

- Joined tables to link launch records with additional data (e.g., rocket details).
- Combined datasets for comprehensive analysis.

Filtering Queries:

- Filtered data to focus on specific launch outcomes (success/failure).
- Applied conditions to extract launches based on criteria like launch date or rocket configuration.

Sorting Queries:

- Sorted data to identify trends or outliers.
- Ordered launches by date or success rate for analysis.

Subqueries:

- Nested queries to calculate derived metrics (e.g., average payload mass per launch site).
- Subqueries used to perform detailed analysis within larger datasets.

GitHub URL: [khayaIBM/4.jupyter-labs-eda-sql-edx_sqlite.ipynb](https://github.com/khayaIBM/4.jupyter-labs-eda-sql-edx_sqlite.ipynb) at main · KhayelihleMfeka/khayaIBM

Build an interactive map with Folium

Map Objects Created

Markers:

- Placed markers to indicate launch sites on the map.
- Each marker represents a specific geographical location where SpaceX launches have occurred.

Circles:

- Added circles around launch sites to visually represent proximity zones.
- Circles help visualize the areas around launch sites that might influence operational decisions.

Lines:

- Drew lines to connect launch sites with their proximities or other relevant locations.
- Lines provide spatial context and connections between different points of interest related to launches.

Github URL:

[khayaIBM/6.lab_jupyter_launch_site_location.ipynb at main · KhayelihleMfeka/khayaIBM](#)

Reasons for Adding Objects

Markers:

- To pinpoint exact launch locations for spatial reference.
- Helps users identify where SpaceX has conducted launches geographically.

Circles:

- Illustrates the potential impact zones around launch sites.
- Provides a visual representation of safety perimeters or operational boundaries.

Lines:

- Shows connections or relationships between launch sites and relevant features.
- Enhances understanding of spatial relationships and dependencies.

Build a dashboard with Plotly Dash

Plots/Graphs Added:

Success Pie Chart:

- Displays the distribution of successful and failed launches.
- Helps visualize the overall success rate and performance trends.

Success-Payload Scatter Plot:

- Shows the relationship between payload mass and launch success.
- Allows users to explore how payload mass influences mission outcomes.

Interactions Added

Launch Site Dropdown:

- Enables users to select specific launch sites for analysis.
- Facilitates filtering and focused exploration based on geographical locations.

Range Slider for Payload:

- Allows users to adjust payload mass ranges dynamically.
- Offers flexibility in examining launch success concerning payload mass variations.
- Github URL: [s khayalBM/spacex-dash-app \(2\).py at main · KhayeliMfeka/khayalBM](https://github.com/KhayeliMfeka/khayalBM/spacex-dash-app)

Reasons for plots:

Success Pie Chart:

- Provides a quick overview of mission success rates.
- Essential for stakeholders to understand overall performance metrics at a glance.

Success-Payload Scatter Plot:

- Helps identify correlations between payload characteristics and launch outcomes.
- Supports decision-making processes related to payload planning and operational strategies.

Interactions Added

Launch Site Dropdown:

- Enhances user experience by focusing analysis on specific launch locations.
- Allows for regional insights and comparisons across different launch sites.

Range Slider for Payload:

- Offers interactive exploration of how payload mass affects mission success.
- Enables detailed analysis and insights into payload-related performance factors.

Predictive analysis (classification)

1. Data Preprocessing:

- Standardized features to ensure all variables contribute equally.
- Split data into training and test sets for model validation.

2. Model Selection:

- Explored multiple classification algorithms: SVM, Decision Trees, and K-Nearest Neighbors (KNN).
- Chose algorithms suitable for binary classification tasks based on project requirements.

3. Hyperparameter Tuning:

- Used GridSearchCV to systematically search for optimal hyperparameters.
- Tuned parameters such as C (SVM), max_depth (Decision Trees), and n_neighbors (KNN).

4. Model Evaluation:

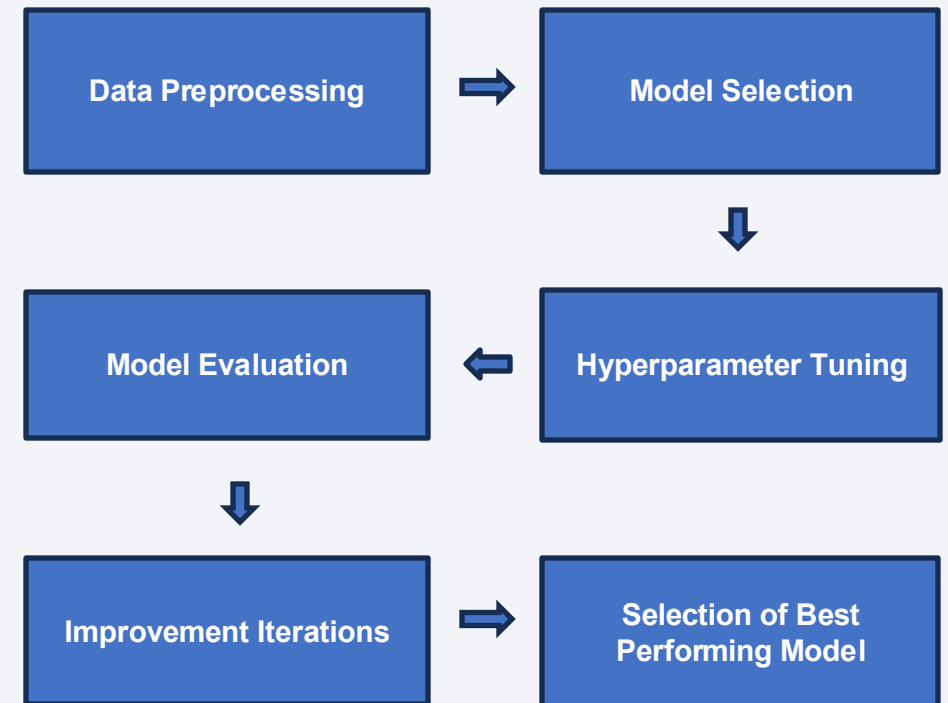
- Evaluated models using cross-validation techniques to ensure robustness and generalizability.
- Utilized metrics like accuracy, precision, recall, and F1-score to assess model performance.

5. Improvement Iterations:

- Iteratively adjusted models based on insights from validation results.
- Fine-tuned hyperparameters to maximize predictive accuracy and reliability.

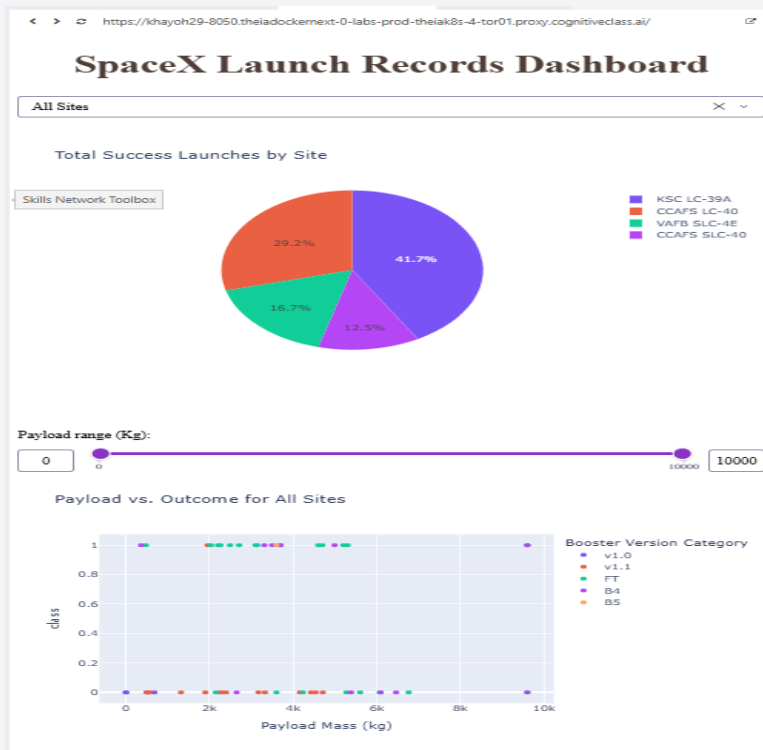
6. Selection of Best Performing Model:

- Identified the model with the highest accuracy on the test set as the best performer.
- Considered both training and test set performance to avoid overfitting and ensure real-world applicability.



Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results



Display the total payload mass carried by boosters launched by NASA (CRS)

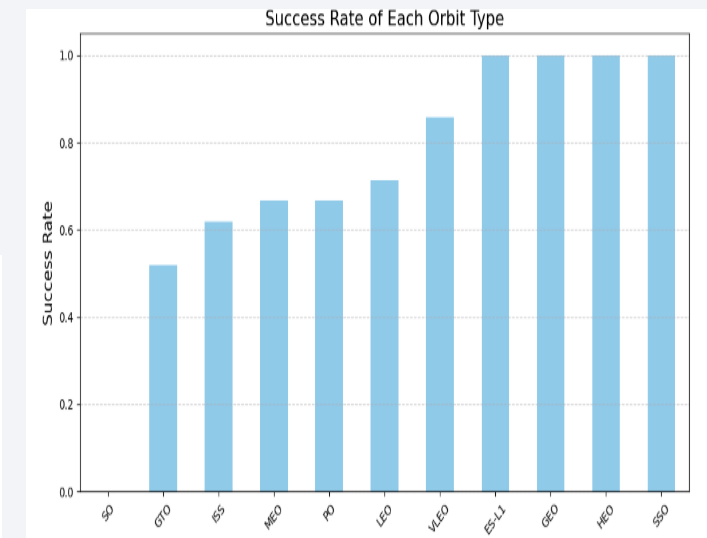
```
%sql SELECT SUM("PAYLOAD_MASS_KG_") FROM SPACESTABLE WHERE "Customer" = 'NASA (CRS)';
```

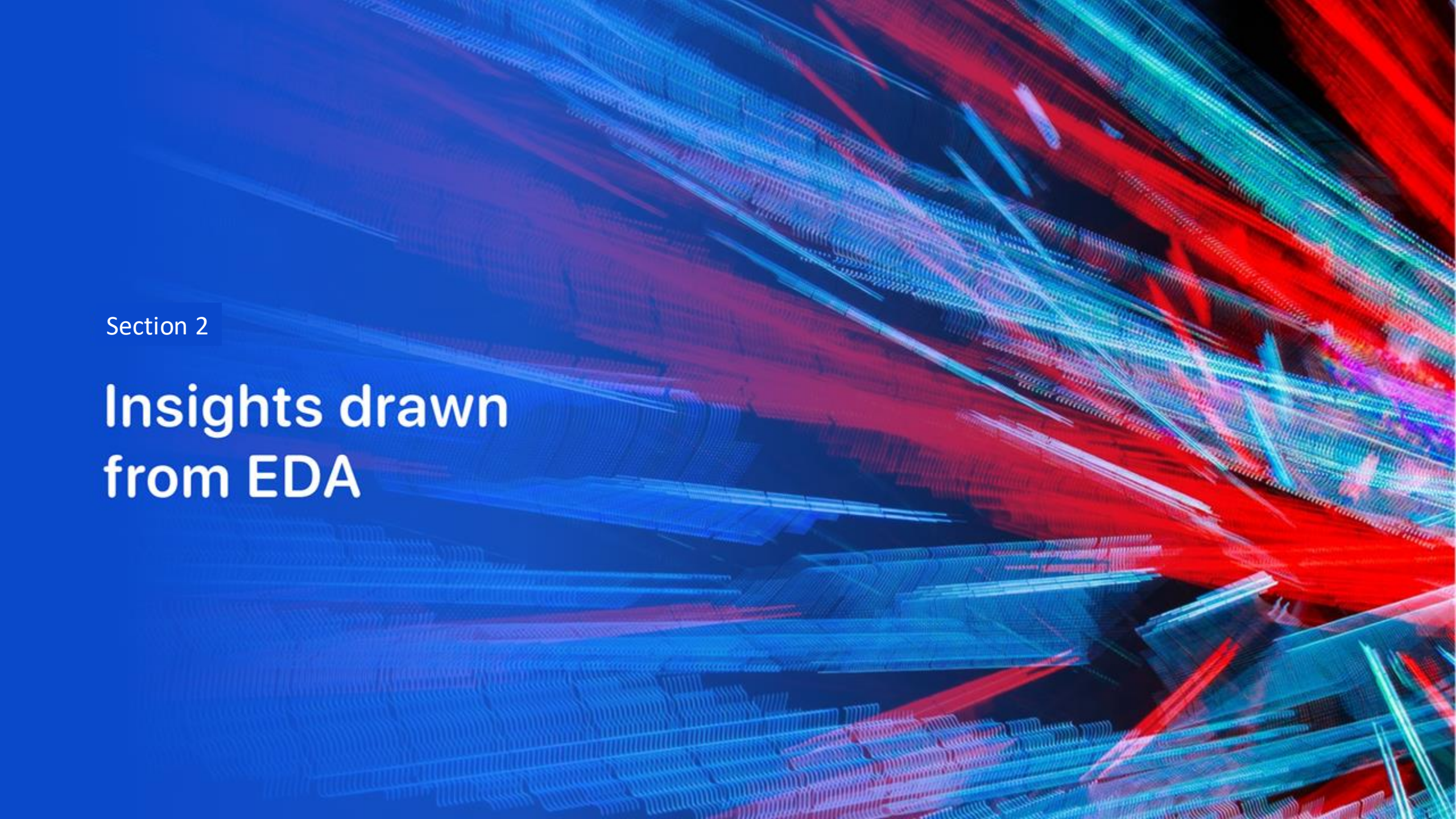
* sqlite:///my_data1.db
Done.

SUM("PAYLOAD_MASS_KG_")
0 45596

1. Logistic Regression: Accuracy on test data: 0.8333
2. Support Vector Machine (SVM): Accuracy on test data: 0.8333
3. Decision Tree: Accuracy on test data: 0.8889
4. K Nearest Neighbors (KNN): Accuracy on test data: 0.8333

The best performing model is the Decision Tree model, which shows the highest accuracy of 0.8889 on the test set.

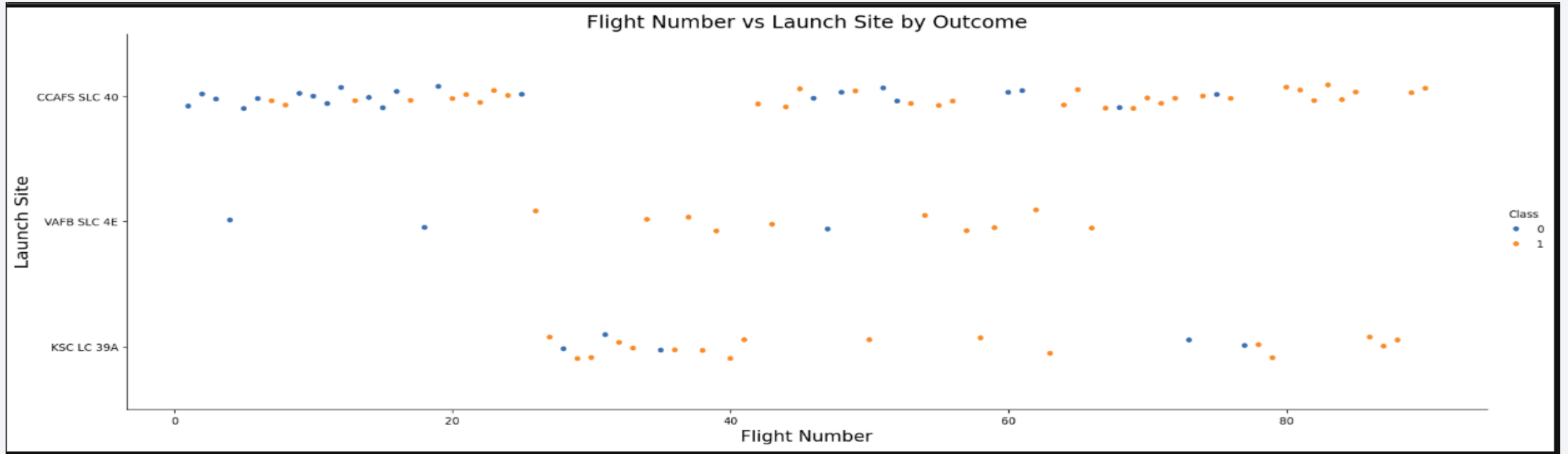




Section 2

Insights drawn from EDA

Flight number vs. launch site

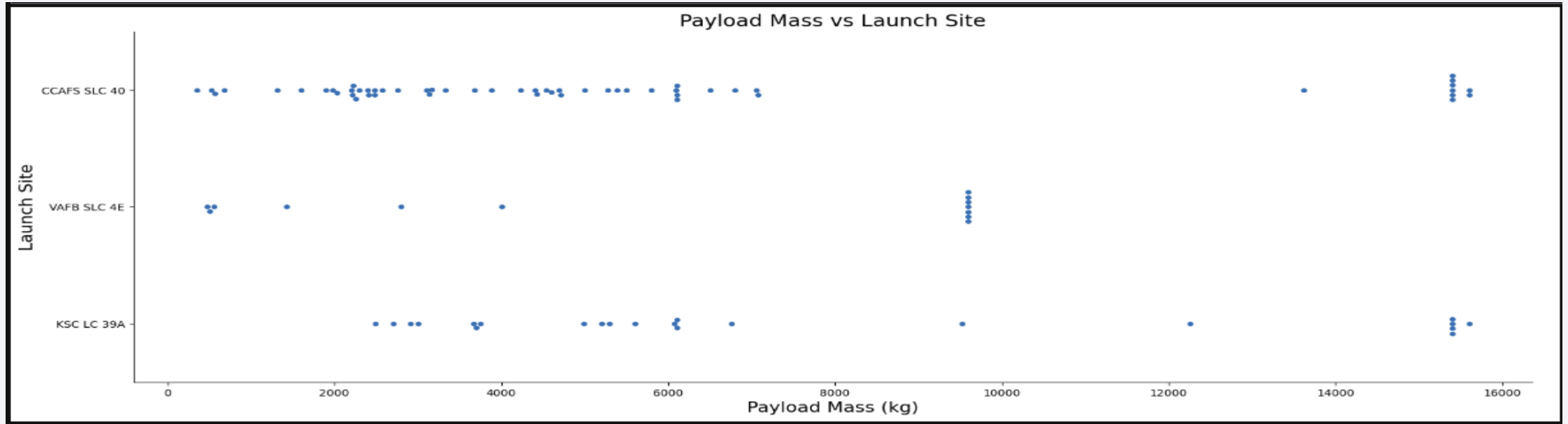


Inferences:

Mixed Outcomes at Major Launch Sites: Both CCAFS SLC-40 and KSC LC-39A exhibit a mix of successful (orange) and unsuccessful (blue) landings, suggesting that factors beyond the launch site itself may play a significant role in determining landing outcomes.

Consistent Activity Across Flight Numbers: Launches are distributed across a broad range of flight numbers at all sites, reflecting consistent activity over time without a discernible trend of increasing or decreasing landing success.

Payload vs. launch site

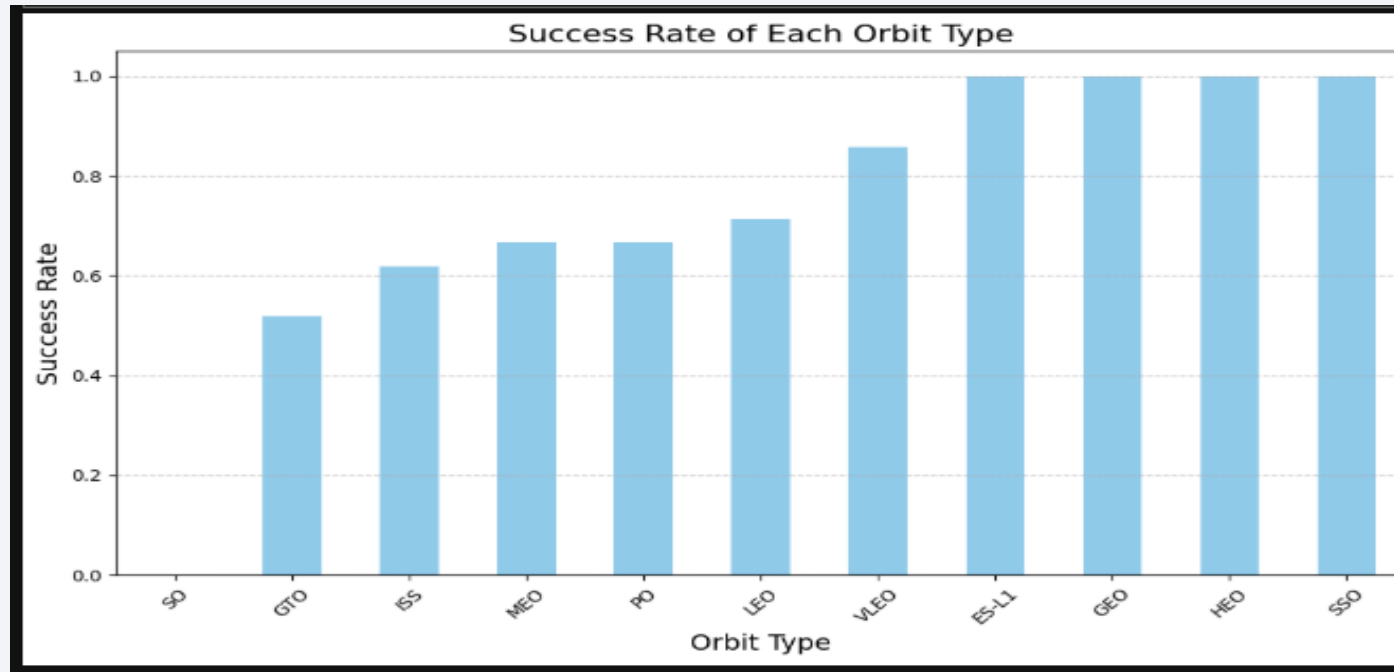


Inferences:

Payload Distribution: Most launches from the CCAFS SLC-40 site involve payloads under 10,000 kg, whereas the VAFB SLC-4E and KSC LC-39A sites accommodate a broader range of payload masses, reflecting more diverse mission profiles.

High-Capacity Launches: The KSC LC-39A site is frequently utilized for missions involving heavier payloads, with multiple launches exceeding 15,000 kg, underscoring its suitability for high-capacity operations.

Success rate vs. orbit type

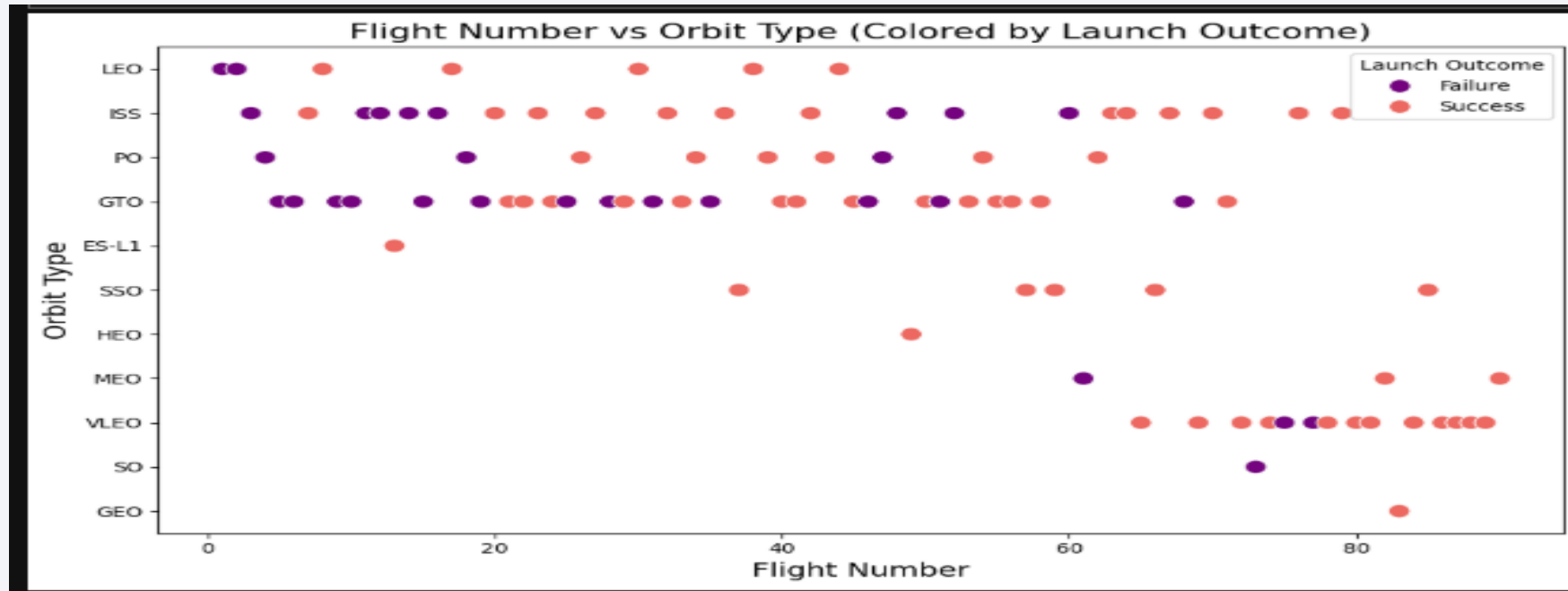


Inferences:

High Success Rates: Missions to VLEO, ES-L1, GEO, HEO, and SSO orbits have achieved a perfect success rate, underscoring the high reliability of these orbital destinations for successful first-stage landings.

Lower Success Rate for GTO: The GTO orbit type exhibits a notably lower success rate compared to other orbital categories, suggesting that missions to this orbit involve greater challenges and operational complexities.

Flight number vs. orbit type

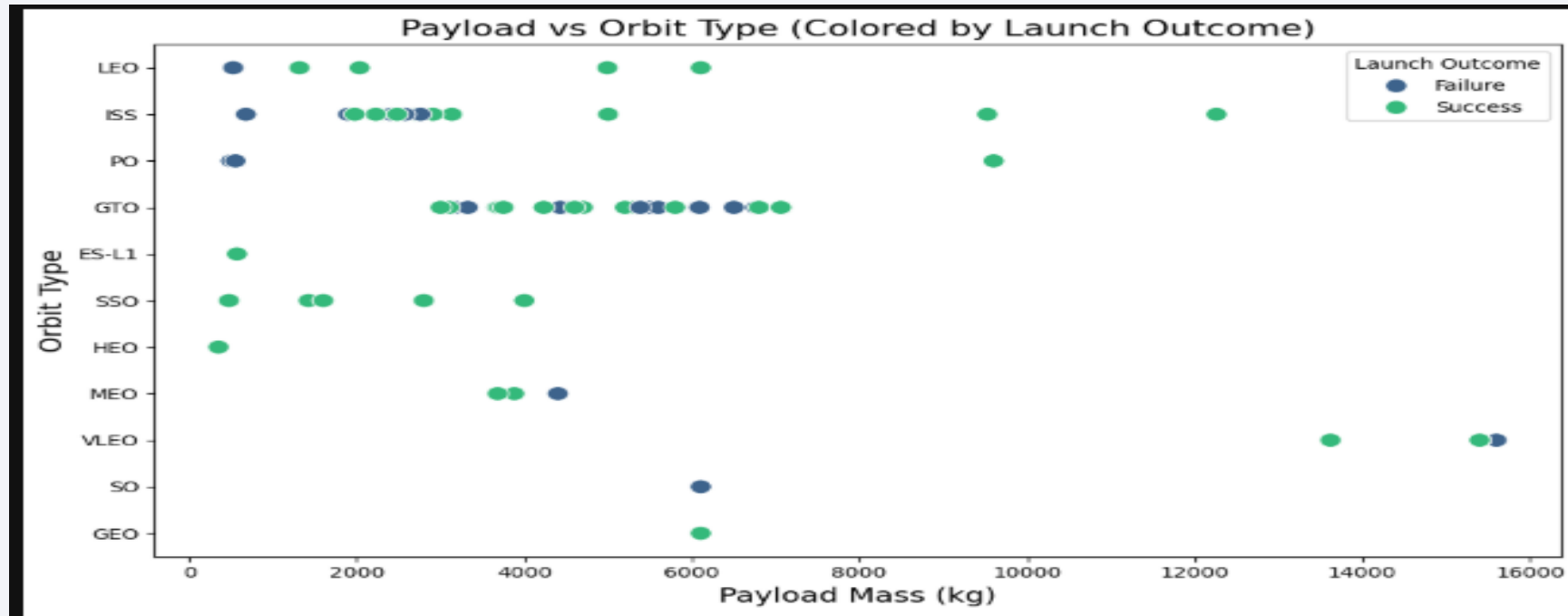


Inferences:

Increased Success Over Time: The success rate of Falcon 9 launches increases markedly with higher flight counts, indicating that accumulated experience and iterative enhancements contribute to improved outcomes.

Orbit-Specific Performance: Early missions to GTO and ISS orbits yielded mixed outcomes, whereas more recent launches to these destinations demonstrate a higher success rate, reflecting advancements in mission planning and execution.

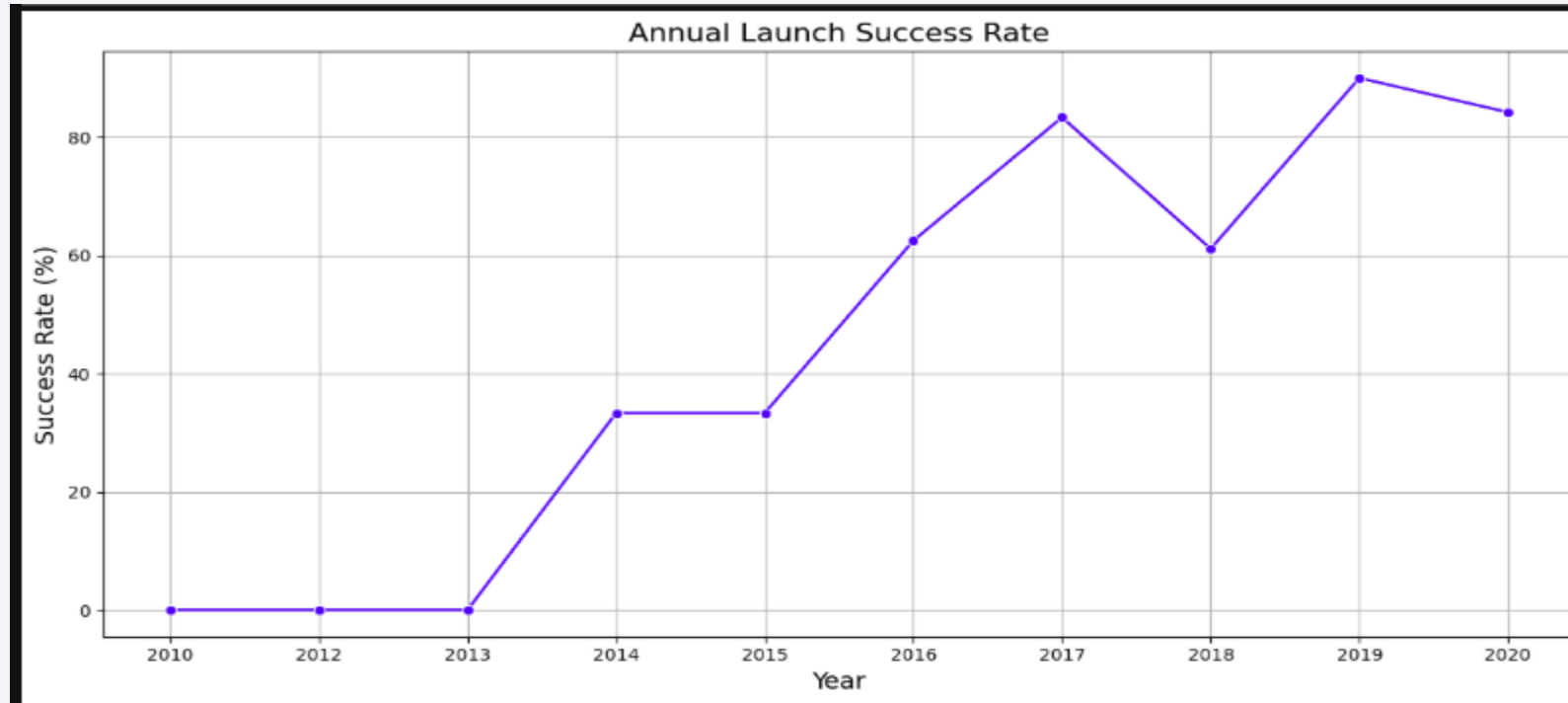
Payload vs. orbit type



Inferences:

Successful landings occur more frequently across all orbit types, particularly for payloads under 6,000 kg. In contrast, payloads exceeding 10,000 kg exhibit a mix of successes and failures, reflecting the increased challenges associated with heavier payloads.

Launch success yearly trend



Inferences:

The annual launch success rate has demonstrated marked improvement since 2013, surpassing 80% by 2020. Despite a temporary decline in 2018, the overall trajectory reflects increasing reliability and sustained success in Falcon 9 launches over time.

All launch site names

```
Display the names of the unique launch sites in the space mission

%sql SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE;

* sqlite:///my_data1.db
Done.
```

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

Inferences:

The unique launch sites are **CCAFS LC-40** , **VAFB SLC-4E**, **KSC LC-39A** and **CCAFS SLC-40**.

Launch site names begin with “KSC”

Display 5 records where launch sites begin with the string 'KSC'

```
%sql SELECT * FROM SPACESTABLE WHERE "Launch_Site" LIKE 'KSC%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
1	2017-03-16	6:00:00	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar	Success	No attempt
2	2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
3	2017-05-01	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	Success	Success (ground pad)
4	2017-05-15	23:21:00	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat	Success	No attempt

Inferences:

The output shows 5 records where sites begin with 'KSC'

Total payload mass

```
Display the total payload mass carried by boosters launched by NASA (CRS)

%sql SELECT SUM("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
* sqlite:///my_data1.db
Done.
SUM("PAYLOAD_MASS_KG_")
0                        45596
```

Inferences:

The total payload mass carried by boosters launched by NASA(CRS) is **45596** kg.

Average payload mass by F9 v1.1

```
Display average payload mass carried by booster version F9 v1.1 ⓘ  
%sql SELECT AVG("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';  
* sqlite:///my_data1.db  
Done.  


| AVG("PAYLOAD_MASS_KG_") |
|-------------------------|
| 2928.4                  |


```

Inferences:

The average payload mass carried by booster version F9 v1.1 is **2928.4** kg.

First successful ground landing date

```
%sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
* sqlite:///my_data1.db
Done.
  MIN("Date")
-----
0  2015-12-22
```

Inferences:

The first successful ground landing date was on the **22nd of December 2015**.

Successful drone ship landing with payload between 4,000 and 6,000

List the names of the boosters which have success in ground pad and have payload mass greater than 4000 but less than 6000

```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" > 4000 AND "PAYLOAD_MASS__KG_" < 6000;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

	Booster_Version
0	F9 FT B1022
1	F9 FT B1026
2	F9 FT B1021.2
3	F9 FT B1031.2

Inferences:

The list of boosters with successful drone ship landing and a payload between 4000 and 6000 kg: **F9 FT B1022**, **F9 FT B1026** , **F9 FT B1021.2** and **F9 FT B1031.2**.

Total number of successful and failed mission outcomes

List the total number of successful and failure mission outcomes

```
%sql SELECT "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

```
Done.
```

	Mission_Outcome	Total
0	Success	98

Inferences:

The number of successful and failed mission outcomes is **98**.

Boosters carried maximum payload

List all the booster_versions that have carried the maximum payload mass. Use a subquery.

```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db
```

```
Done.
```

	Booster_Version
0	F9 B5 B1048.4
1	F9 B5 B1049.4
2	F9 B5 B1051.3
3	F9 B5 B1056.4
4	F9 B5 B1048.5
5	F9 B5 B1051.4
6	F9 B5 B1049.5
7	F9 B5 B1060.2
8	F9 B5 B1058.3
9	F9 B5 B1051.6
10	F9 B5 B1060.3
11	F9 B5 B1049.7

2015 launch records

List the records which will display the month names, succesful landing_outcomes in ground pad ,booster versions, launch_site for the months in year 2017 1

Note: SQLite does not support monthnames. So you need to use substr(Date,6,2) for month, substr(Date,9,2) for date, substr(Date,0,5),='2017' for year.

```
%%sql
```

```
SELECT
CASE
WHEN substr("Date", 6, 2) = '01' THEN 'January'
WHEN substr("Date", 6, 2) = '02' THEN 'February'
WHEN substr("Date", 6, 2) = '03' THEN 'March'
WHEN substr("Date", 6, 2) = '04' THEN 'April'
WHEN substr("Date", 6, 2) = '05' THEN 'May'
WHEN substr("Date", 6, 2) = '06' THEN 'June'
WHEN substr("Date", 6, 2) = '07' THEN 'July'
WHEN substr("Date", 6, 2) = '08' THEN 'August'
WHEN substr("Date", 6, 2) = '09' THEN 'September'
WHEN substr("Date", 6, 2) = '10' THEN 'October'
WHEN substr("Date", 6, 2) = '11' THEN 'November'
WHEN substr("Date", 6, 2) = '12' THEN 'December'
ELSE 'Unknown'
END AS "Month_Name",
"Mission_Outcome",
"Booster_Version",
"Launch_Site"
FROM
SPACEXTABLE
WHERE
substr("Date", 0, 5) = '2015';
```

```
* sqlite:///my_data1.db
Done.
```

	Month_Name	Mission_Outcome	Booster_Version	Launch_Site
0	January	Success	F9 v1.1 B1012	CCAFS LC-40
1	February	Success	F9 v1.1 B1013	CCAFS LC-40
2	March	Success	F9 v1.1 B1014	CCAFS LC-40
3	April	Success	F9 v1.1 B1015	CCAFS LC-40
4	April	Success	F9 v1.1 B1016	CCAFS LC-40
5	June	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40
6	December	Success	F9 FT B1019	CCAFS LC-40

Rank landing outcomes between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%%sql
SELECT
    "Landing_Outcome",
    COUNT(*) AS "Count"
FROM
    SPACEXTABLE
WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    COUNT(*) DESC;
```

```
* sqlite:///my_data1.db
Done.
```

	Landing_Outcome	Count
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

1. Mark all launch sites on a map

Are all launch sites in proximity to the Equator line?

- Not all launch sites are close to equator.
- Vandenberg Air Force Base (VAFB SLC 4E) launch site is located at a latitude of 34.63 , which is further from the Equator compared to the other sites in Florida.

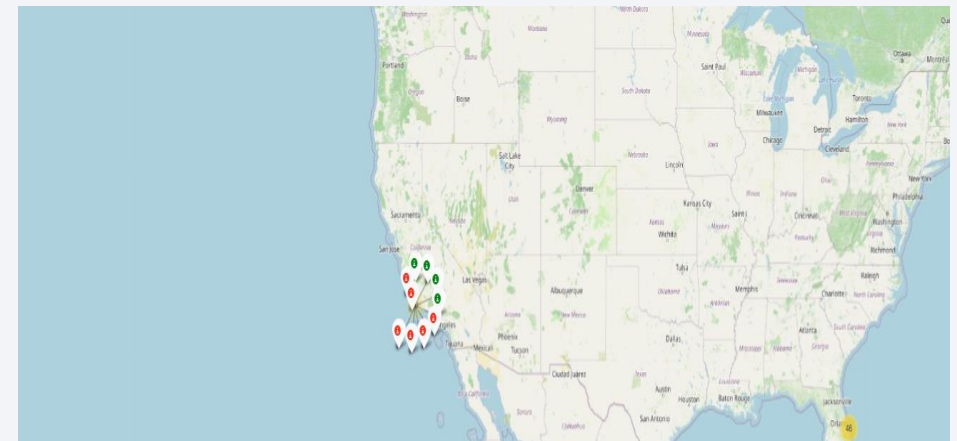
Are all launch sites in very close proximity to the coast?

- Yes, all launch sites are near the coast.
- The Cape Canaveral sites (CCAFS LC-40 and CCAFS SLC-40) and Kennedy Space Center (KSC LC-39A) are near the coast in Florida.
- • Vandenberg Air Force Base (VAFB SLC-4E) is also near the coast in California.



<Mark the success/failed launches for each site on the map>

- This plot depicts the distance between the CCAFS SLC-40 launch site and the nearest coastline, measured at approximately 0.51 kilometers as indicated by the marker. The PolyLine highlights the straight-line measurement, underscoring the site's close proximity to the coast. Such positioning is typical of launch facilities, as it supports over-water flight paths and safe recovery operations, thereby reducing risks to populated areas.
- For instance, in the provided screenshot, 10 launches are shown from the VAFB SLC-4E site, with 6 red markers and 4 green markers. This color-coding enables quick identification of success rates and other categorical distinctions. The red markers denote unsuccessful launches, while the green markers represent successful ones, offering immediate visual feedback on the site's performance.



Calculate the distances between a launch site to its proximities

This plot illustrates the distance between the CCAFS SLC-40 launch site and the nearest coastline, calculated at approximately 0.51 kilometers as indicated by the marker. The added PolyLine clearly depicts the straight-line measurement, emphasizing the site's close proximity to the coast. Such positioning is characteristic of launch facilities, as it enables over-water flight paths and safe recovery operations, thereby minimizing risks to populated areas.

Inferences:

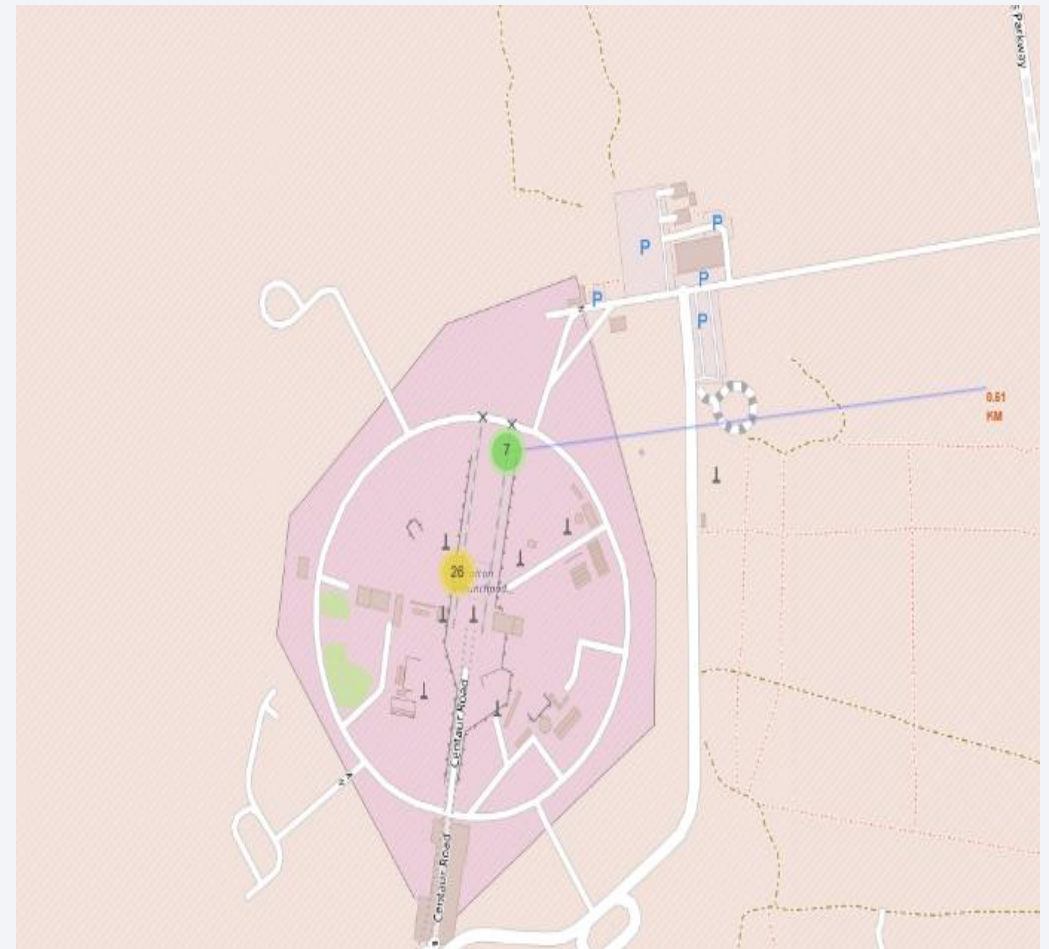
The first successful ground landing date was on the **22nd of December 2015**.

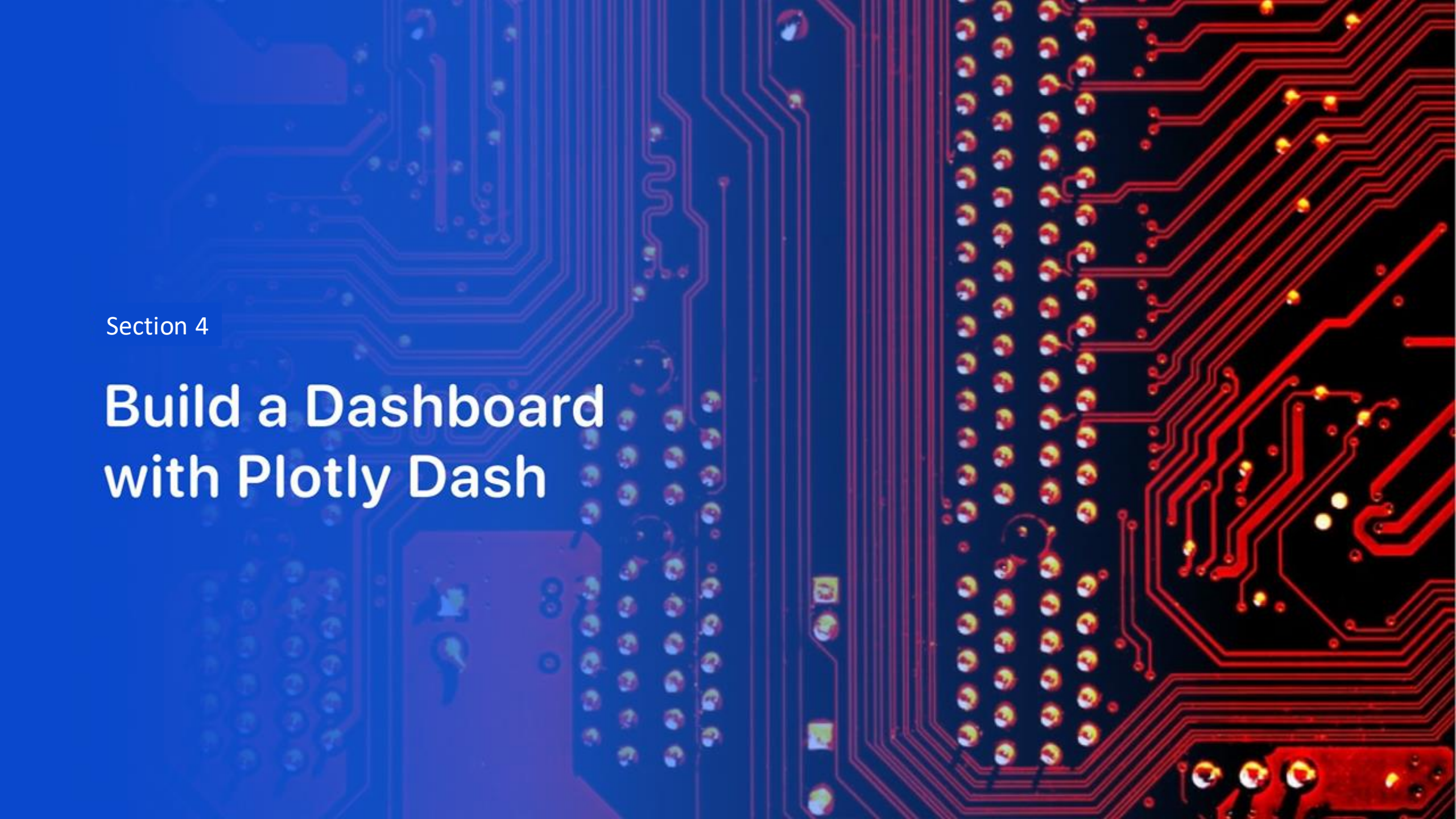
```
[30]: coastline_lat = 28.56367
      coastline_lon = -80.57163

      # Example launch site coordinates (replace with actual launch site coordinates)
      launch_site_lat = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Lat'].values[0]
      launch_site_lon = launch_sites_df.loc[launch_sites_df['Launch Site'] == 'CCAFS SLC-40', 'Long'].values[0]

      # Calculate distance using the calculate_distance function
      distance_coastline = calculate_distance(launch_site_lat, launch_site_lon, coastline_lat, coastline_lon)

      print(f"Distance from launch site to closest coastline: {distance_coastline} km")
      Distance from launch site to closest coastline: 0.5097439631188212 km
```



The background of the slide is a close-up, artistic photograph of a printed circuit board (PCB). The board is dark, and the intricate circuit traces are highlighted in a vibrant, glowing red. Numerous small, circular components, likely solder joints or micro-components, are visible along the traces, some of which also appear to be glowing. The lighting creates a sense of depth and technological sophistication.

Section 4

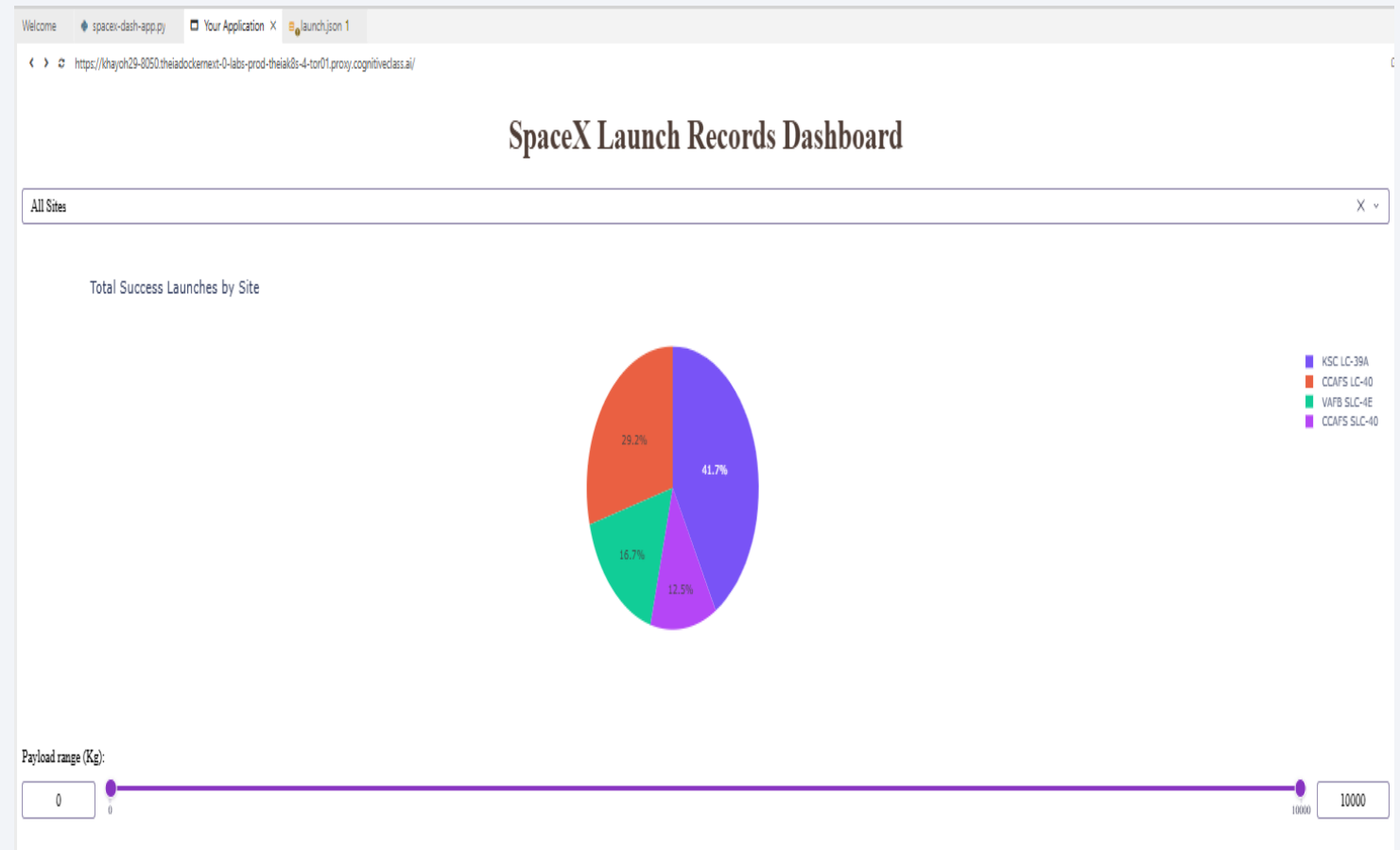
Build a Dashboard with Plotly Dash

launch success count for all sites (Pie Chart)

Findings and Insights:

- KSC LC-39A: 41.7%
- CCAFS LC-40: 29.2%
- VAFB SLC-4E: 16.7%
- CCAFS SLC-40: 12.5%

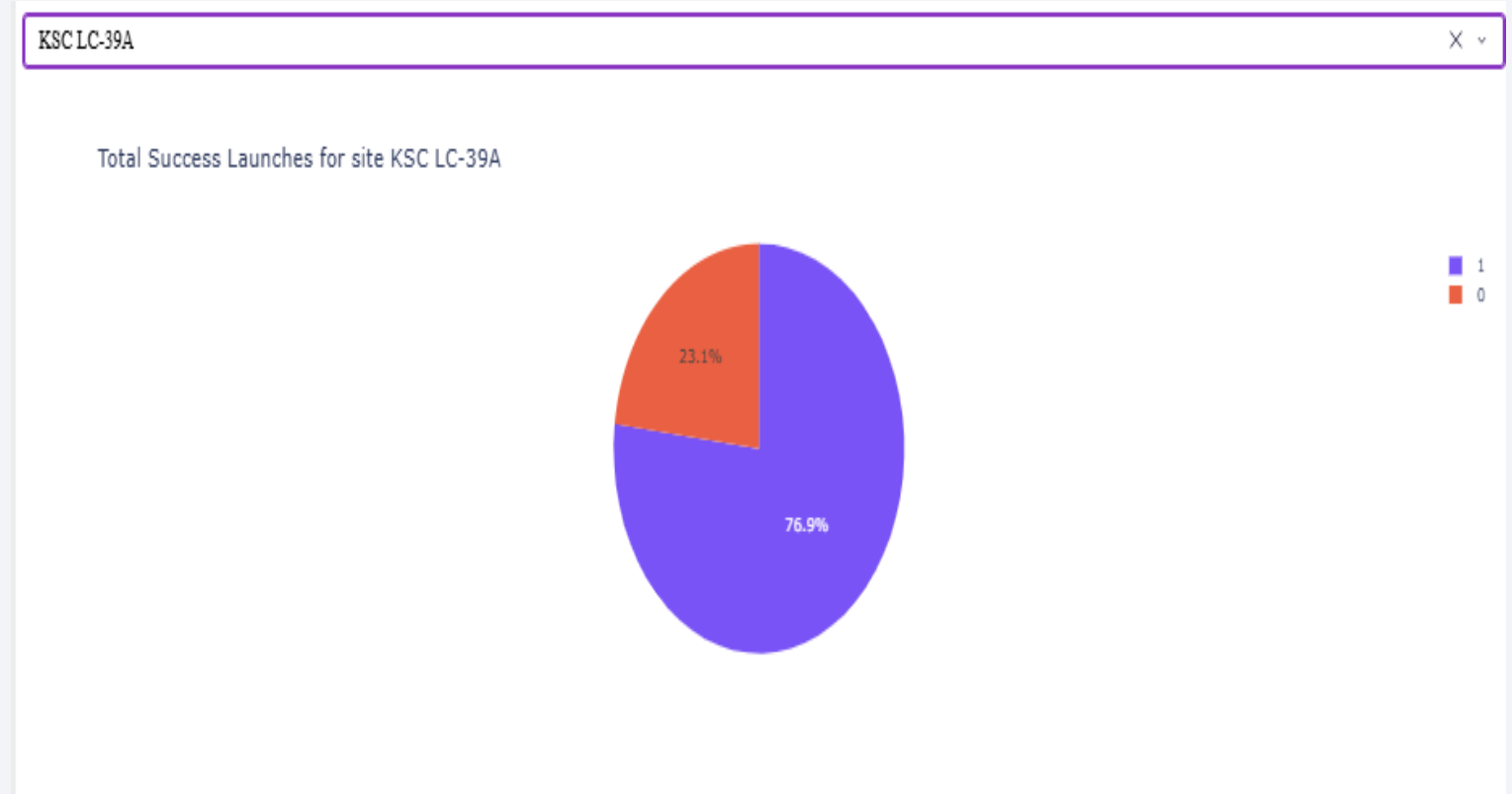
The KSC LC-39A launch site records the highest number of successful missions, accounting for 41.7% of total successes. This performance underscores its reliability and establishes it as one of SpaceX's most effective launch sites.



Launch Site with highest success ratio (pie chart)

Findings and Insights:

- The substantial number of successful launches from KSC LC-39A underscores its reliability and effectiveness as a primary launch site.
- KSC LC-39A:
 - Class 1 (Successful Launches): 76.9%
 - Class 0 (Unsuccessful Launches): 23.1%
- The high success rate (76.9%) for Class 1 launches underscores the effectiveness and reliability of the KSC LC-39A site..



SpaceX Launch Dashboard Insights

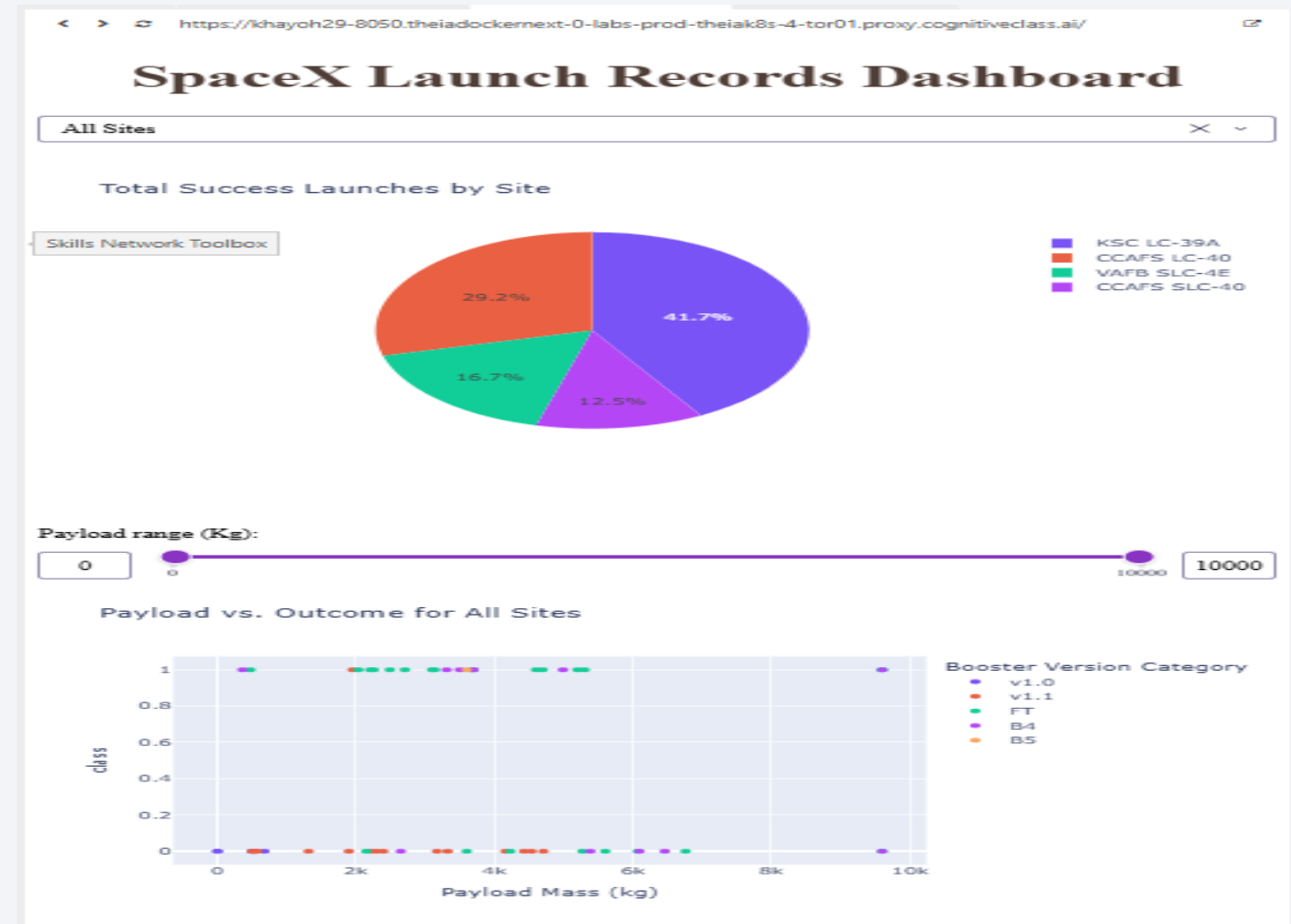
Findings and Insights:

Launch Site Success Rates:

- KSC LC-39A, has the highest success rate with 41.7% of successful launches.
- This suggests that KSC LC-39A is the most reliable launch site among the ones analyzed.
- Other sites like CCAFS LC-40, VAFB SLC 4E, and CCAFS SLC-40 have lower success rates, indicating variability in launch success across different sites.

Booster Version Performance:

- The “FT” booster version emerges as the most frequently utilized and demonstrates a consistently high success rate across a wide range of payload masses.
- The “v1.0” booster version has comparatively fewer launches and warrants further analysis to fully evaluate its performance.
- Overall, the analysis of booster versions does not reveal a definitive trend linking higher payload masses to reduced success rates, indicating that payload mass alone is not a determining factor in launch outcomes.



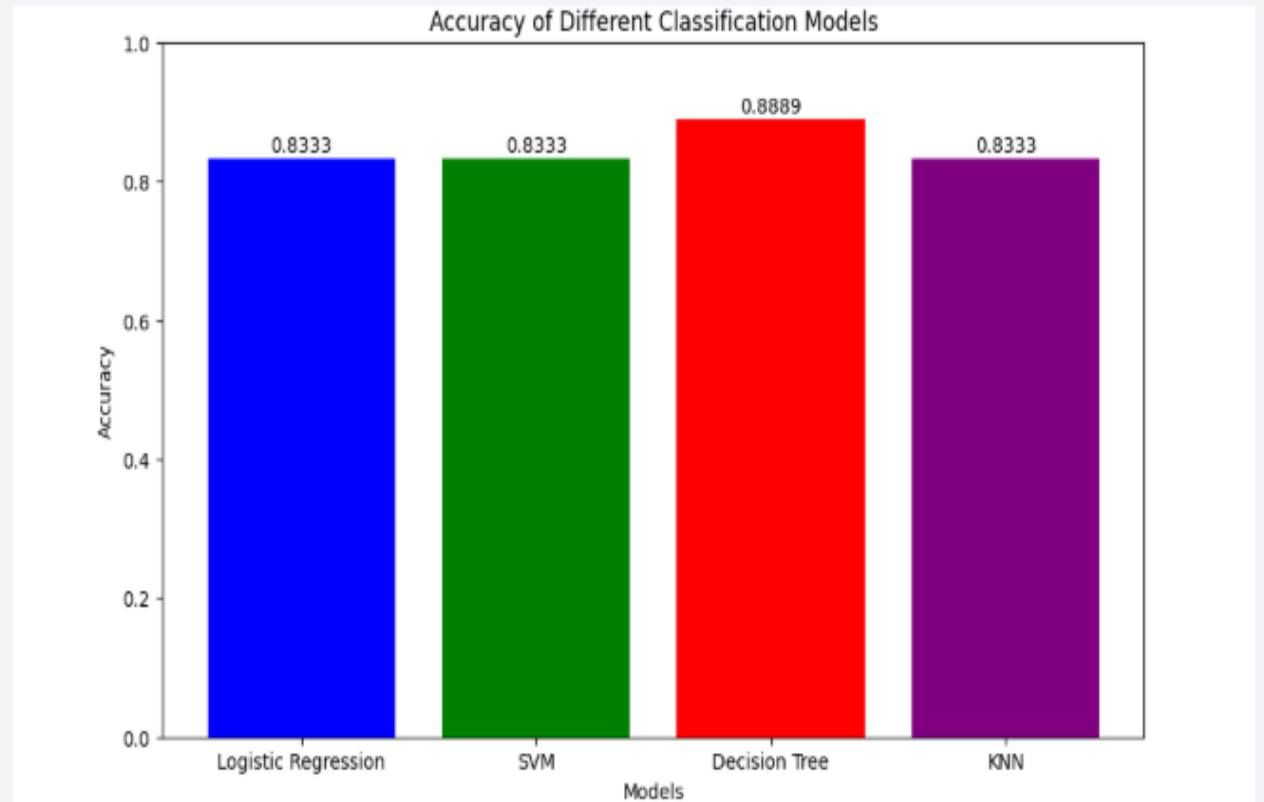


Section 5

Predictive Analysis (Classification)

Classification accuracy

- Based on the evaluation results, the Decision Tree model achieved the highest classification accuracy on the test dataset, with a score of 0.8889. This indicates that the Decision Tree is better suited for this dataset compared to Logistic Regression, Support Vector Machine, and K Nearest Neighbors, each of which reached an accuracy of 0.8333.



Confusion matrix

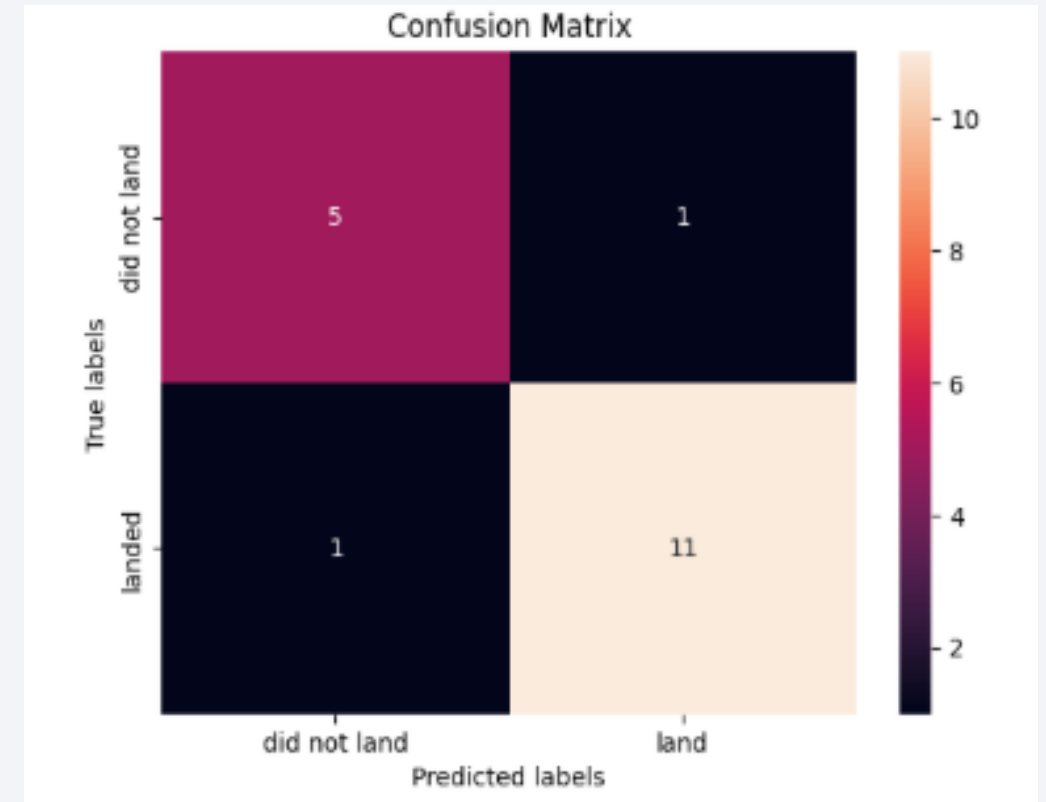
Explanation :

High Accuracy: The model achieved a high accuracy score of 88.89%, with a significant number of true positives and true negatives, demonstrating its effectiveness in predicting Falcon 9 first stage landings.

Manageable False Negatives: The presence of a false negative indicates that the model failed to recognize one actual successful landing, instead predicting it as a failure. This misclassification highlights a limitation in the model's sensitivity, as it overlooked a genuine positive outcome. In aerospace operations, such an error could lead to underestimating the reliability of the landing system, potentially affecting cost estimations and readiness planning.

Manageable False Positives: Although the model generated one false positive, this outcome is considerably less critical than false negatives in aerospace applications. An overestimation of success merely results in additional preparation, which is far easier to accommodate than the risks posed by insufficient readiness. Consequently, the model's overall performance remains highly acceptable and well-suited for practical implementation.

Balanced Performance: The model exhibits balanced performance, with a slight inclination toward predicting successful landings. This bias aligns with the operational priorities of the aerospace industry, where precise identification of successful recoveries is vital for accurate cost estimation, efficient resource allocation, and long-term strategic planning.



Conclusions

- Point 1: Our analysis revealed that the "KSC LC-39A" launch site has the highest success rate among all sites, accounting for 41.7% of successful launches. This indicates that this site might have optimal conditions or processes that contribute to a higher success rate.
- Point 2: The scatter plot analysis revealed that the "FT" booster version consistently achieves a high success rate across diverse payload masses, underscoring its reliability and robustness relative to other booster variants. This finding suggests that future missions could benefit from prioritizing the use of this booster version to enhance overall success rates.
- Point 3: No definitive correlation was identified between increased payload mass and reduced landing success, suggesting that other variables—such as launch site conditions and booster configurations—exert a more substantial influence on launch outcomes.
- Point 4: Interactive visualizations developed with Folium and Plotly Dash offered valuable insights into the geographical and operational dynamics of SpaceX launches. By leveraging these tools, the analysis provided a deeper understanding of the data, equipping stakeholders with comprehensive visual analytics to support informed decision-making.

In summary, our predictive analysis and interactive visualizations have illuminated the critical factors underlying SpaceX's launch success while establishing a robust framework for future evaluation and strategic decision-making within the aerospace sector. The insights derived from this study not only enhance the optimization of launch strategies but also contribute meaningfully to the continued advancement and success of reusable rocket technology.

Thank you!

